DATASHEET

# ChArUco calibration target, composite

ChArUco combines a checkerboard with ArUco markers in the white squares. Markers identify each corner, so the board can be partly outside the image or partly occluded and still contribute corners — which is exactly where lens distortion is strongest. Native in OpenCV (`cv2.aruco.CharucoBoard`), supported by MATLAB and most multi-camera toolchains.

## 1 Ordering

| Part number | Size (W × H) | Squares (rows × cols) | Square / marker | Dictionary | Corners | Markers |
|---|---|---|---|---|---|---|
| CT-CHA-P-200150-{DICT} | 200 × 150 mm | 5 × 7 | 25 / 18 mm | see below | 24 | 17 |
| CT-CHA-P-300200-{DICT} | 300 × 200 mm | 7 × 10 | 25 / 18 mm | | 54 | 35 |
| CT-CHA-P-400300-{DICT} | 400 × 300 mm | 9 × 12 | 30 / 22 mm | | 88 | 54 |
| CT-CHA-P-600400-{DICT} | 600 × 400 mm | 9 × 14 | 40 / 30 mm | | 104 | 63 |
| CT-CHA-P-800600-{DICT} | 800 × 600 mm | 11 × 15 | 50 / 37 mm | | 140 | 82 |

## 2 Key specifications

ACCURACY

±0.3 mm + 0.2 mm/m (any corner to nominal grid, at 20 °C)

SUBSTRATE

Aluminium / LDPE / aluminium composite, 6 mm

FLATNESS

≤ 0.5 mm per metre, board supported on three points

Unless stated otherwise, dimensions are nominal at 20 °C.

## 1 Specifications · geometry

| FEATURE | SPECIFICATION |
|---|---|
| Feature position tolerance | ±0.3 mm + 0.2 mm/m (any corner to nominal grid, at 20 °C) |
| Square pitch, mean | ±0.05 mm |
| Orthogonality | ±0.02° |
| Marker-to-square ratio | 0.74 |
| Edge radius | R6 mm |

## 2 Specifications · physical

| FEATURE | SPECIFICATION |
|---|---|
| Substrate | Aluminium / LDPE / aluminium composite, 6 mm |
| Weight | 6.6 kg/m² (300 × 200: ≈ 0.4 kg · 800 × 600: ≈ 3.2 kg) |
| Pattern | Direct UV print, matte, black on white; optical density ≥ 1.6 |
| Flatness | ≤ 0.5 mm per metre, board supported on three points |
| Rigidity (E·J) | ≈ 3800 kN·cm²/m |
| Mounting | 4 × M4 threaded inserts on the rear (300 × 200 and larger); ¼″ fixture ACC-A-FIX-14 clamps any size |

## 3 Specifications · environment

| FEATURE | SPECIFICATION |
|---|---|
| Operating temperature | −50 to +80 °C |
| Thermal expansion | ≈ 24 µm/(m·°C) |
| Storage | −20 to +60 °C, dry |
| Outdoor lifetime | up to 3 years (UV-stable print) |
| Cleaning | water or ≤ 70 % isopropyl alcohol, soft cloth |

## 4 Deliverables

| INCLUDED | | SERVICE | |
|---|---|---|---|
| Inspection report | CT-IR-, per board, dimensional + flatness + visual | Lead time | Made to order · ships in 2–3 weeks |
| Files | `.opencv.json`, `.kalibr.yaml` (checkerboard mode), `.ros2.yaml`, `.halcon.descr`, `.dxf` | Options | ILAC-MRA certificate issued by an accredited third-party laboratory, available as SVC-X-CERT-ILAC (+1–3 weeks) · logo print (-L) · hard case (ACC-A-CSE) |
| Documents | this datasheet, CT-UM-GEN user manual, CT-UM-CHA-P quick start | | |

## 5 Compliance & customs

| TOPIC | STATEMENT |
|---|---|
| RoHS | RoHS-compliant materials. |
| HS code | `9031 80` Not a toy. not for food contact. |

## 6 Revision history

| REV | DATE | CHANGE |
|---|---|---|
| A | `2026-10` | First issue, planning values |

## 7 Disclaimer

Values are nominal unless stated as measured; per-unit values are in the inspection report.